AI-POWERED COGNITIVE SCREENING & DEMENTIA RISK ANALYSIS PLATFORM

Project Final Report

Project ID: 25-26J-439

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BSc (Hons) in Information Technology Specializing in Software Engineering

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DECLARATION

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The above candidates have carried out these research theses for the Degree of Bachelor of Science (Honors) in Information Technology (Specializing in Software Engineering) under my supervision.

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Signature of co-supervisor

(Ms. Kaushalya Rajapakse)

Abstract

Early identification of dementia risk is essential for timely clinical intervention; however, traditional screening approaches often rely on manual assessments and limited behavioral indicators. This study presents CogniCare, a multimodal AI-assisted cognitive screening platform designed to support clinicians through integrated digital assessments. The system enables users to complete lifestyle profiling, cognitive and functional evaluations, handwriting analysis, and guided speech tasks through a unified platform. Collected health and behavioral data are analyzed using machine learning and standardized assessment methods to estimate dementia risk, while the results are integrated through rule-based analysis to generate an initial screening outcome. Speech and handwriting modules further analyze behavioral patterns associated with cognitive decline, and all outputs are consolidated into a comprehensive report available through a clinician dashboard. Guided by DSM-5 diagnostic criteria, physicians determine whether additional neurological evaluation is required, and MRI scans can be further analyzed using deep learning-based dementia stage classification to assist clinical decision-making. The proposed platform demonstrates the potential of integrating multimodal digital assessments with clinical workflows for scalable dementia risk screening while ensuring that final diagnostic decisions remain under medical supervision. Experimental evaluation demonstrates strong performance across system modules, where the wearable-based CatBoost model achieved an AUC of 0.989 and 94.9% accuracy, the speech hybrid model achieved an F1-score of 0.82, the handwriting model achieved an F1-score of 0.870, and the MRI ensemble model achieved 93.75% accuracy with a Macro F1-score of 0.9606.

Keywords

Dementia Screening, Multimodal Health Analysis, Cognitive Assessment, Machine Learning, Speech Analysis, Handwriting Analysis, MRI Classification, Clinical Decision Support.

Acknowledgement

We would like to express our sincere gratitude to our project supervisor for their continuous guidance, support, and invaluable feedback throughout the development of this research. Their expertise and encouragement played a significant role in shaping the direction and successful completion of this project.

We also extend our sincere appreciation to our co-supervisor and all academic staff for their valuable insights, constructive suggestions, and support provided during different stages of this study. Their contributions have greatly enhanced the quality of this work.

We would like to thank all team members for their collaboration, dedication, and teamwork in developing the multimodal CogniCare system. The successful integration of components such as wearable health monitoring, cognitive assessment, speech and handwriting analysis, and MRI-based classification would not have been possible without the collective effort of the team.

We also acknowledge the support of external technologies and resources, including open-source machine learning frameworks, wearable platforms such as Google Fit, and publicly available datasets, which contributed significantly to the implementation of this system.

Finally, we express our heartfelt gratitude to our families for their continuous encouragement, patience, and support throughout our academic journey.

This research would not have been possible without the contribution and support of all these individuals and resources.

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1. Introduction

1.1. Background & Literature Survey

Dementia is a progressive neurodegenerative disorder characterized by a decline in cognitive abilities, including memory, language, reasoning, and executive functioning. It significantly impacts an individual's ability to perform daily activities and maintain independence. According to global health statistics, dementia is one of the leading causes of disability and dependency among older adults, with millions of new cases reported annually. The increasing ageing population worldwide has further accelerated the prevalence of dementia, making it a critical public health concern.

Early detection of dementia plays a vital role in slowing disease progression and improving patient outcomes. Identifying cognitive decline at its initial stages allows clinicians to implement therapeutic interventions, lifestyle modifications, and monitoring strategies that can significantly delay the severity of symptoms. However, conventional diagnostic approaches are often limited in their ability to detect early-stage dementia.

Traditional screening methods primarily rely on standardized cognitive assessments such as the Addenbrooke's Cognitive Examination (ACE-III) [1] and Mini-Mental State Examination (MMSE). These tests evaluate memory, attention, language, and visuospatial abilities. While effective, these assessments are typically conducted in controlled clinical environments and require trained healthcare professionals. This dependency makes them less accessible and not suitable for continuous or large-scale screening.

In addition to cognitive tests, neuroimaging techniques such as Magnetic Resonance Imaging (MRI) and Computed Tomography (CT) scans are widely used to identify structural abnormalities in the brain associated with dementia. These imaging methods provide detailed insights into brain atrophy and neural degeneration. Recent advancements in deep learning have enabled automated analysis of MRI data, improving classification accuracy in detecting different stages of dementia.

Machine learning has become a key enabler in modern healthcare applications, particularly in disease prediction and classification. Various machine learning algorithms such as Random Forest, Support Vector Machines (SVM), and Gradient Boosting techniques have been applied to dementia detection. These models can learn complex patterns from large datasets and provide predictive insights based on extracted features.

Beyond clinical and imaging data, researchers have increasingly focused on behavioral biomarkers as indicators of cognitive decline. Behavioral data is non-invasive, cost-effective, and can be collected in real-world environments. Among these, speech analysis has emerged as a promising approach. Cognitive decline often manifests in speech through reduced fluency, increased pauses, limited vocabulary, and difficulty in forming coherent sentences. Acoustic features such as pitch variation, speech rate, and energy levels provide valuable insights into neurological health.

Similarly, handwriting analysis has been explored as a digital biomarker for detecting dementia. Writing involves a combination of cognitive processing, motor coordination, and spatial awareness. Changes in handwriting patterns, including stroke dynamics, writing speed, and pressure variations, can indicate early cognitive impairment. Machine learning models trained on handwriting data have demonstrated the ability to distinguish between healthy individuals and those with cognitive disorders.

Wearable technologies further enhance the ability to monitor cognitive health continuously. Devices capable of tracking physiological signals such as heart rate, sleep patterns, and physical activity provide valuable data related to lifestyle and behavioral patterns. Studies have shown that irregular sleep, reduced physical activity, and abnormal physiological signals are associated with increased dementia risk.

Despite these advancements, most existing systems are designed around a single modality. For instance, some models focus solely on speech analysis, while others rely on MRI imaging or cognitive test results. However, dementia is inherently a multidimensional condition influenced by a combination of behavioral, physiological, and neurological factors.

The integration of multiple data sources into a unified analytical framework has the potential to significantly improve detection accuracy. Multimodal systems can capture a broader range of indicators, enabling more comprehensive analysis of cognitive health. This research builds upon this concept by proposing a multimodal AI-powered system that integrates speech, handwriting, wearable health data, and neuroimaging to support early dementia detection.

1.2. Research Gap

Although existing research has demonstrated the effectiveness of machine learning in dementia detection, several gaps remain that limit the practical application and accuracy of current systems.

One of the primary gaps is the reliance on single-modality approaches. Most studies focus on specific types of data, such as speech signals, handwriting dynamics, or MRI scans. While these approaches provide valuable insights, they fail to capture the complex and multifaceted nature of cognitive decline. Dementia affects multiple domains simultaneously, and analyzing a single modality may lead to incomplete or biased predictions.

Another significant gap is the lack of integration between different data sources. Behavioral data, physiological signals, and neuroimaging information are often analyzed independently. This fragmented approach prevents the development of a holistic understanding of cognitive health. Integrating these diverse data types into a unified system remains a challenging yet essential task.

Scalability is another major concern in existing systems. Traditional diagnostic methods require clinical environments and trained professionals, making them difficult to implement on a large scale. Many machine learning models are developed using controlled datasets, which may not reflect real-world conditions. As a result, their applicability in real-world scenarios is limited.

Furthermore, many machine learning models lack interpretability. Deep learning models, although highly accurate, often function as black boxes, making it difficult for clinicians to understand how predictions are generated. In healthcare applications, interpretability is crucial for building trust and ensuring reliable decision-making.

Another gap is the limited use of continuous monitoring systems. Most current approaches rely on periodic assessments rather than continuous data collection. Wearable technologies offer the potential for real-time monitoring, but their integration into cognitive screening systems is still limited.

Additionally, there is a lack of comprehensive platforms that combine multiple screening methods into a single system. Existing solutions are often isolated applications that focus on a specific aspect of dementia detection. A unified platform that integrates multiple modules can significantly improve usability and effectiveness.

This research addresses these gaps by proposing a multimodal system that integrates behavioral, physiological, and neuroimaging data into a single framework. The system is designed to be scalable, interpretable, and capable of supporting continuous monitoring, thereby overcoming the limitations of existing approaches.

Research Work	**Multi-Domain Data (Lifestyle + Clinical + Wearable)**	**Cognitive & Functional Integration (ACE-III / IADL)**	**Real-Time / Wearable Integration**	**Explainability / Interpretable Output**
Research A (L., 2020)	✓ (Lifestyle + clinical)	✗	✗	✗
Research B (K., 2021)	✓ (Wearable + activity)	✗	✓	✗
Research C (M., 2022)	✓ (Clinical + demographic)	Partial (Cognitive only)	✗	✗
Research D (S., 2023)	Partial (Lifestyle only)	✗	✗	✗
Research E (R., 2024)	✓ (Multi-source data)	✗	✓	Partial (basic feature importance)
Proposed CogniCare System	✓	✓	✓	✓

Table 1 Smart Health Monitoring & Dementia Risk Analysis

Feature / Capability	Balestrino et al. (2024)	Mitra & Rehman (2024)	D'Alessandro et al. (2024)	Cilia et al. (2018)	Ye et al. (2023)	Proposed System (CogniCare)
Handwriting-based cognitive assessment	✓	✓	✓	✓	✗	✓
Use of machine learning models	✗	✓	✓	✗	✗	✓
High classification accuracy	✓	✓	✓	✗	✗	✓
Use of standardized dataset (DARWIN)	✗	✓	✗	✓	✗	✓
Dynamic feature extraction	✓	✓	✓	✓	✗	✓
Advanced feature engineering	✗	✓	✓	✗	✗	✓
Reduced feature set for efficiency	✗	✗	✗	✗	✗	✓
Real-time processing capability	✗	✗	✗	✗	✗	✓
Browser-based data collection	✗	✗	✗	✗	✗	✓
Device independence	✗	✗	✗	✗	✓	✓
Interpretability (Explainable AI)	✗	✗	✗	✗	✗	✓
Scalability for real-world deployment	✗	✗	✗	✗	✗	✓
Usability in non-clinical environments	✗	✗	✗	✗	✓	✓
Lightweight implementation	✗	✗	✗	✗	✓	✓
End-to-end automated system	✗	✓	✗	✗	✗	✓

Table 2 Comparative Analysis of Existing Handwriting-Based Dementia Screening Approaches and the Proposed System

Feature / Capability	Zhang et al. (2021)	Luz et al. (2021)	Genewein et al. (2022)	Tóth et al. (2020)	Ferreira et al. (2019)	Proposed System (CogniCare – Speech Module)
Guided multi-task speech assessment (Self Introduction, Morning Recall, Picture Description)	✕	✕	✓	✕	✕	✓
Automatic speech-to-text transcription	✓	✓	✕	✓	✕	✓
Acoustic feature extraction (e.g., duration, tempo, ZCR, RMS energy)	✓	✓	✓	✓	✕	✓
Linguistic feature extraction (e.g., word count, lexical variation, narrative structure)	✕	✓	✕	✓	✕	✓
Use of machine learning / deep learning models	✓	✓	✓	✓	✕	✓
High classification accuracy	✓	✓	✓	✕	✕	✓
Use of standardized dataset (e.g., Pitt, DementiaBank)	✕	✓	✓	✕	✕	✓
Real-time dementia risk prediction	✕	✕	✕	✕	✕	✓

Table 3 Comparative Analysis of Speech-Based Dementia Screening Systems and the Proposed CogniCare Speech Module

Research Work	Multi-Stage Classification (CDR)	Explainability (XAI)	Dementia Specific
Research A (X., 2025)	✓	✗	Partial (Alzheimer's-focused)

Research B (C.I., 2023)	✗	✗	Partial (AD detection)
Research C (O.B., 2014)	✗	✗	Partial (AD vs control)
Research D (I.O., 2016)	✗	✗	Partial (MCI→AD focus)
Research E (M., 2023)	✗	✗	✗ (brain tumor task)
Research F (E.M., 2024)	✗	✗	✗ (brain tumor task)
Proposed Project	✓	✓	✓

Table 4 Comparative Analysis of MRI-Based Dementia Classification Approaches

Feature / Capability	**Luz et al. (2020)**	**Araújo et al. (2020)**	**Albers et al. (2022)**	**Bäckström et al. (2020)**	**Ye et al. (2023)**	**Proposed System (CogniCare)**
Multimodal approach	✗	✗	✗	✗	✗	✓
Handwriting-based assessment	✗	✓	✗	✗	✗	✓
Speech-based assessment	✓	✗	✗	✗	✗	✓
Wearable / lifestyle data	✗	✗	✓	✗	✓	✓
MRI / neuroimaging analysis	✗	✗	✗	✓	✗	✓
Use of machine learning / deep learning	✓	✓	✓	✓	✗	✓
High classification performance	✓	✓	✓	✓	✗	✓
Real-time / continuous monitoring	✗	✗	✓	✗	✗	✓

Browser-based data collection	✗	✗	✗	✗	✗	✓
Device independence	✗	✗	✗	✗	✗	✓
Interpretability (Explainable AI)	✗	✗	✗	✗	✗	✓
Scalability for real-world deployment	✗	✗	✗	✗	✗	✓

Table 5 Comparative Analysis of Existing Dementia Screening Approaches and the Proposed CogniCare System

1.3. Research Problem

Despite significant advancements in machine learning and digital health technologies, early detection of dementia remains a complex and unresolved challenge. Current diagnostic methods are primarily based on clinical evaluations and imaging techniques, which are not only resource-intensive but also limited in their ability to detect early-stage cognitive decline.

One of the major challenges is the inability of existing systems to capture subtle behavioral changes that occur in the early stages of dementia. These changes are often not noticeable through traditional assessments but can be detected through analysis of speech, handwriting, and daily activity patterns.

Another challenge is the lack of integration between different types of data. Behavioral indicators, physiological signals, and neuroimaging data are often analyzed separately, leading to fragmented insights. This lack of integration reduces the effectiveness of existing systems in providing accurate and comprehensive assessments.

Scalability is also a critical issue. Many current systems are designed for clinical use and require specialized equipment and expertise. This limits their accessibility, particularly in remote or resource-constrained environments.

Furthermore, the lack of interpretability in machine learning models poses a significant barrier to adoption in healthcare. Clinicians require transparent and explainable systems that provide insights into how predictions are made.

Based on these challenges, the research problem can be defined as:

How can a scalable, interpretable, and accurate multimodal system be developed to enable early detection of dementia by integrating speech, handwriting, physiological, and neuroimaging data?

1.4. Research Objectives

Main Objective

The main objective of this research is to design and develop an AI-powered multimodal cognitive screening system that integrates speech analysis, handwriting evaluation, wearable health monitoring, and MRI-based classification to support early detection of dementia.

Specific Objectives

- To analyze and extract meaningful features from speech data, including acoustic and linguistic characteristics associated with cognitive decline
- To develop a handwriting analysis module that captures motor and cognitive patterns using digital writing behavior
- To integrate wearable health data to assess lifestyle and physiological factors related to dementia risk
- To implement deep learning models for MRI-based dementia stage classification
- To combine outputs from multiple modalities into a unified decision-support system
- To evaluate the performance of the system using machine learning metrics such as accuracy, precision, recall, and F1-score

1.5. Business Objectives

In addition to the technical and research-oriented goals, this study also considers the practical and commercial potential of the proposed system. The business objective of this research is to develop a scalable, accessible, and cost-effective digital health solution that can support early dementia screening in both clinical and non-clinical environments.

One of the primary business objectives is to reduce the dependency on traditional, resource-intensive diagnostic methods by providing an automated platform that can perform preliminary cognitive assessments. By leveraging widely available technologies such as web-based applications and wearable devices, the system aims to increase accessibility to dementia screening, particularly in regions with limited healthcare infrastructure.

Another key objective is to enable remote and continuous monitoring of cognitive health. Unlike conventional clinical assessments that are performed at specific intervals, the proposed system allows users to perform speech tasks, handwriting assessments, and lifestyle monitoring from their own environment. This supports early identification of cognitive decline and facilitates timely medical intervention, thereby improving patient outcomes.

The platform is also designed to support healthcare professionals by providing interpretable reports and decision-support insights. By consolidating outputs from multiple modalities, including speech, handwriting, wearable health data, and MRI analysis—the system assists clinicians in making informed decisions while maintaining transparency and reliability.

From a commercialization perspective, the system has the potential to be deployed as a Software-as-a-Service (SaaS) solution, where healthcare providers, clinics, and wellness platforms can integrate the system into their existing services. Additionally, the platform can be extended into mobile health applications, enabling large-scale adoption and continuous user engagement.

Furthermore, the system emphasizes data security, privacy, and ethical considerations, ensuring compliance with healthcare data regulations. This is essential for building trust among users and facilitating real-world deployment.

Overall, the business objective of this research is to bridge the gap between advanced AI-based cognitive screening technologies and practical healthcare applications by delivering a scalable, user-friendly, and clinically supportive solution for early dementia detection.

2. Methodology

2.1. Methodology

2.1.1 Introduction

The methodology of this research focuses on the design and development of **CogniCare**, an AI-assisted multimodal dementia screening platform. The system integrates smart health monitoring, cognitive and functional assessments, handwriting analysis, guided speech-task analysis, and optional MRI-based stage classification to support early dementia risk identification.

From a technical perspective, CogniCare follows a **system-based experimental machine learning approach**, where each assessment module operates independently and produces modality-specific outputs. These outputs are later combined into a clinician-facing screening report. This design was selected to improve interpretability, scalability, and clinical usability, rather than depending on a single black-box prediction model.

The methodology chapter is organized as follows. Section 2.2 presents the overall system overview and workflow. Section 2.3 explains the core components of CogniCare, including smart health monitoring, handwriting analysis, speech-based screening, and MRI analysis. Section 2.4 discusses the development process and project management approach. Section 2.5 explains the commercialization aspects of the product, while Section 2.6 presents the testing and implementation strategy used to validate the system.

By following this structure, the methodology demonstrates how CogniCare was systematically designed as a practical, interpretable, and clinically supportive dementia screening platform.

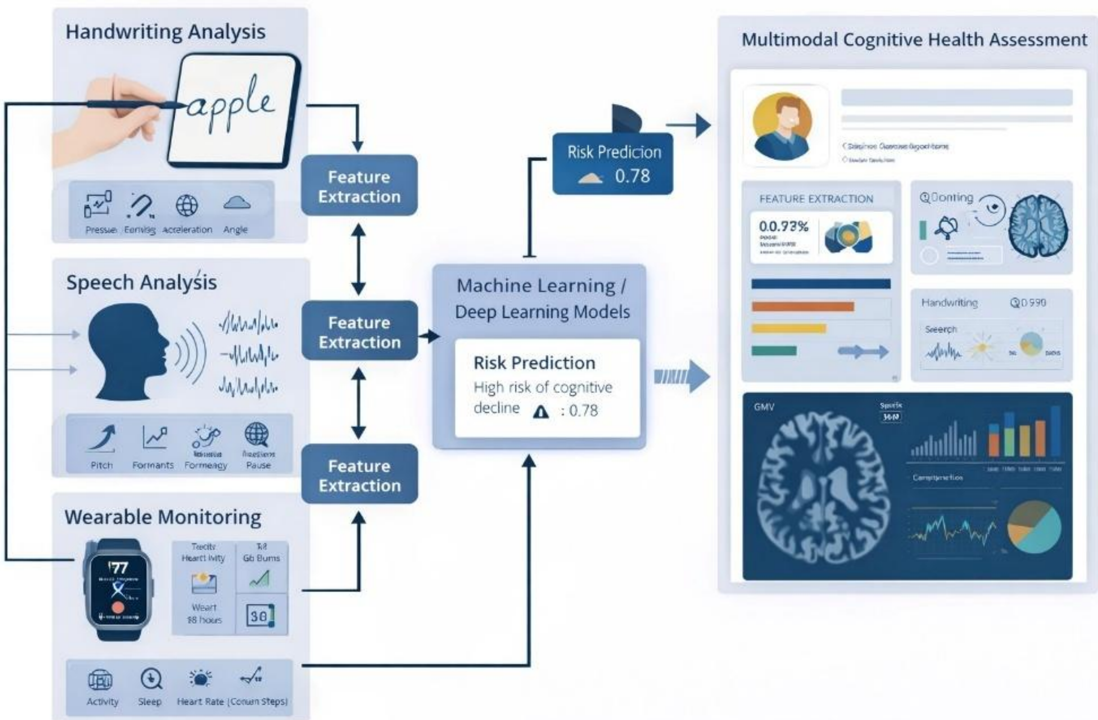


Figure 1 End-to-End Multimodal Dementia Screening Pipeline with Module-Level Processing and Report Consolidation

2.1.2 System Overview

The CogniCare system is designed as a **multimodal, AI-powered cognitive screening platform** that enables early detection of dementia risk through the integration of behavioral, physiological, and neuroimaging data. The system follows a **modular and layered architecture**, allowing each component to independently process specific data modalities while contributing to a unified decision-making framework.

As illustrated in *Figure 1*, the overall system architecture consists of multiple interconnected layers, including the **input layer, preprocessing layer, machine learning layer, decision layer, fusion layer, and output visualization layer**. This structured design ensures scalability, flexibility, and efficient handling of heterogeneous data sources.

The **input layer** collects diverse data types from users, including demographic information, lifestyle factors, chronic health conditions, wearable device data, cognitive assessment scores, and functional ability measures. Behavioral inputs such as handwriting samples and speech recordings are captured through browser-based interfaces, while MRI data is optionally provided for advanced clinical analysis. This multimodal data acquisition enables the system to capture a comprehensive representation of an individual's cognitive health.

Following data collection, the **processing and preprocessing layers** perform essential data preparation tasks such as cleaning, missing value handling, normalization, and feature engineering. These steps ensure data consistency and improve the quality of input for machine learning models. Feature extraction techniques are applied to derive meaningful indicators, including statistical summaries, behavioral patterns, and domain-specific features relevant to each modality.

The **machine learning layer** applies specialized models tailored to each data type. The lifestyle and physiological data are analyzed using a CatBoost gradient boosting classifier, optimized through hyperparameter tuning techniques such as Optuna and validated using stratified cross-validation. The handwriting module utilizes Random Forest and XGBoost models trained on dynamic writing features such as stroke speed, acceleration, and timing patterns. The speech module employs a hybrid approach combining deep learning (Wav2Vec2-XLSR-300M) with acoustic and linguistic feature-based models. The MRI module leverages convolutional neural networks, including DenseNet-121, EfficientNet-B0, and ResNet-50, combined through an ensemble mechanism to improve classification robustness.

The **decision layer** generates intermediate predictions for each modality, such as dementia risk probabilities, cognitive states, and impairment levels. These outputs are then passed to the **fusion layer**, where a rule-based integration mechanism combines results from multiple sources, including cognitive assessment scores (ACE-III), functional ability (IADL), wearable risk predictions, and behavioral analysis outputs. This late fusion approach ensures that each modality contributes independently while maintaining interpretability.

The system incorporates an **explainability layer**, particularly in the handwriting and MRI modules, using techniques such as SHAP (SHapley Additive Explanations) and Grad-CAM++. These methods provide insights into feature importance and highlight the factors influencing model predictions, thereby enhancing transparency and clinician trust.

The **final output layer** presents the aggregated dementia risk classification in clinically meaningful categories, such as Normal, Mild, Moderate, and Severe. These results are displayed through a **clinician dashboard**, which provides comprehensive reports, visualizations, and justification for predictions. The system also supports further evaluation, such as MRI-based analysis, when required.

Each module operates within a **distributed and scalable framework**, enabling deployment in both clinical and non-clinical environments. The use of browser-based interfaces ensures device independence, while API-driven communication supports integration with external systems such as wearable health platforms.

Overall, the CogniCare system addresses the limitations of traditional dementia screening approaches by providing a **comprehensive, interpretable, and scalable solution** that integrates multiple data modalities into a unified platform for early cognitive risk assessment.

2.1.3 Component Overview

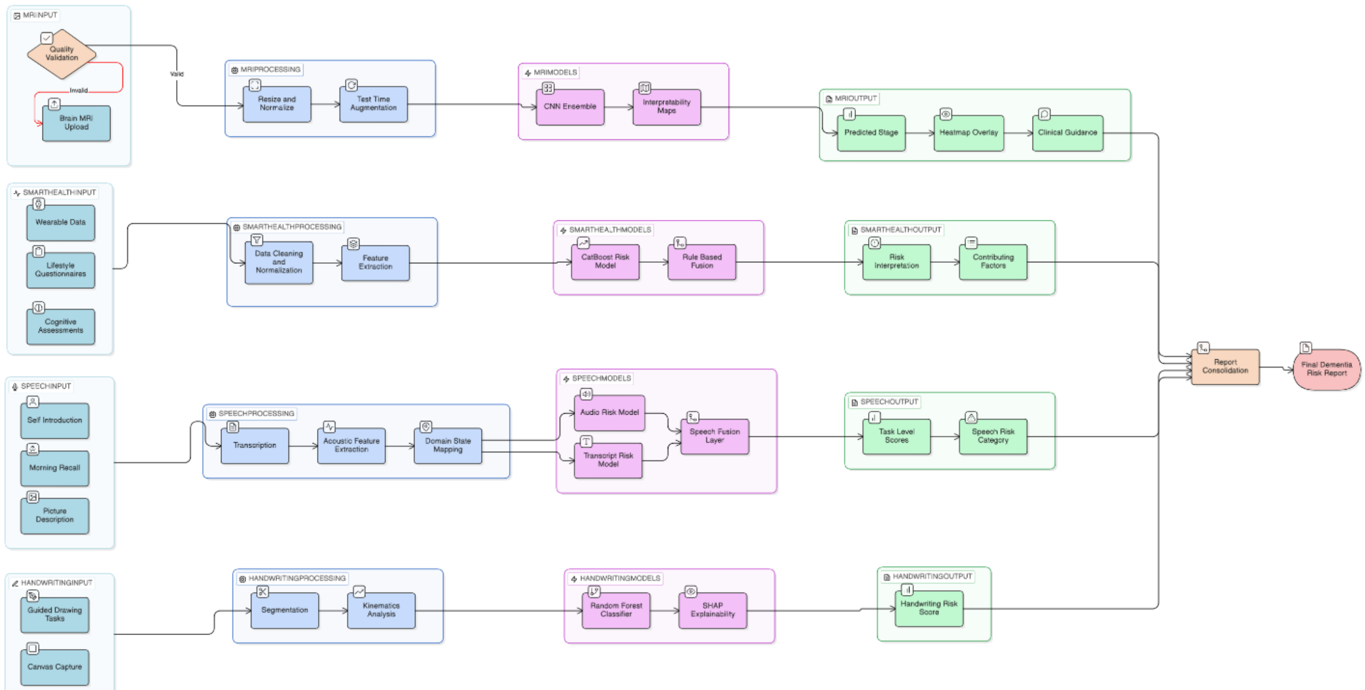


Figure 2 Detailed Multimodal Data Processing and Decision Fusion Workflow of the CogniCare System

1. Smart Health Monitoring and Dementia Risk Analysis

This component collects demographic, lifestyle, physiological, cognitive, and functional data to estimate dementia risk. User inputs include age, gender, education level, smoking habits, alcohol use, sleep behavior, chronic conditions, diabetes status, and depression indicators. Wearable data such as heart rate, step count, and physical activity are obtained through Google Fit integration.

The module also includes ACE-III [1]aligned cognitive assessment and Lawton IADL-based functional assessment. A CatBoost model is used to estimate dementia risk from health and lifestyle features. The results are combined with cognitive and functional states through a rule-based mechanism to generate interpretable risk categories.

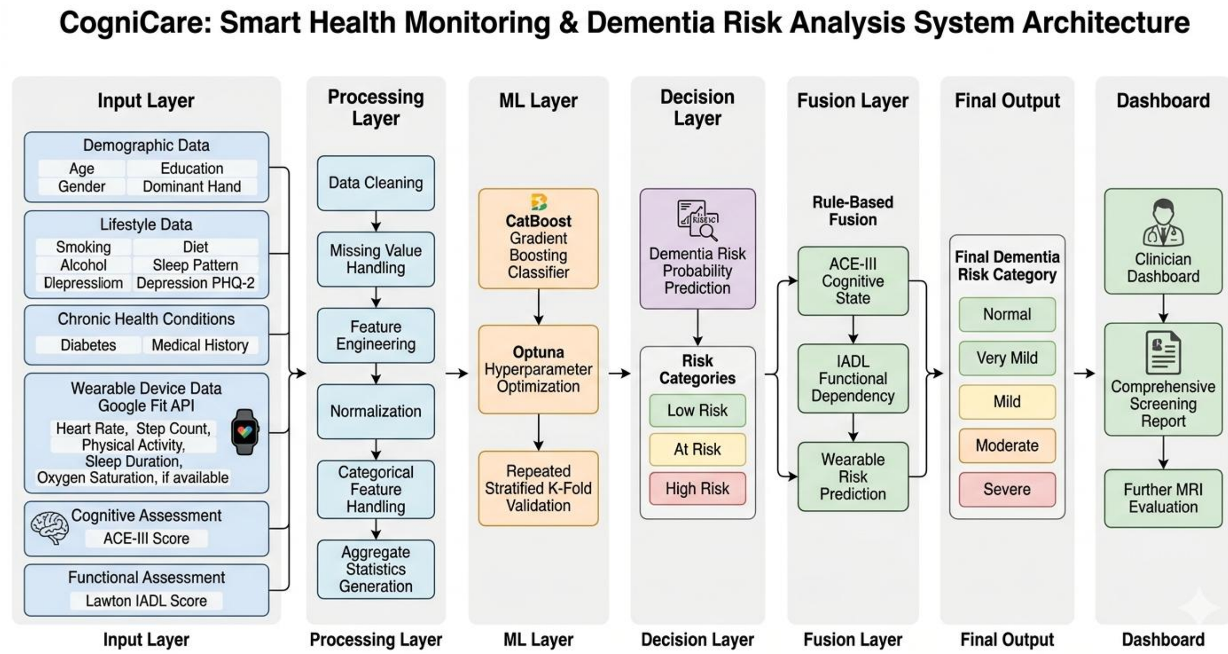


Figure 3 Smart Health Monitoring & Deentia Risk Analysis System Architecture

2. Handwriting-Based Dementia Screening

The handwriting module captures digital writing behavior through a browser-based canvas. During handwriting or drawing tasks, the system records x-y coordinates and timestamps. These data are used to extract behavioral features such as stroke velocity, acceleration, movement duration, and inter-stroke timing.

Random Forest and XGBoost models are used to classify handwriting patterns related to cognitive impairment. To support interpretability, SHAP is applied to identify the handwriting features that contribute most to the prediction. This module provides a lightweight, non-invasive behavioral screening method.

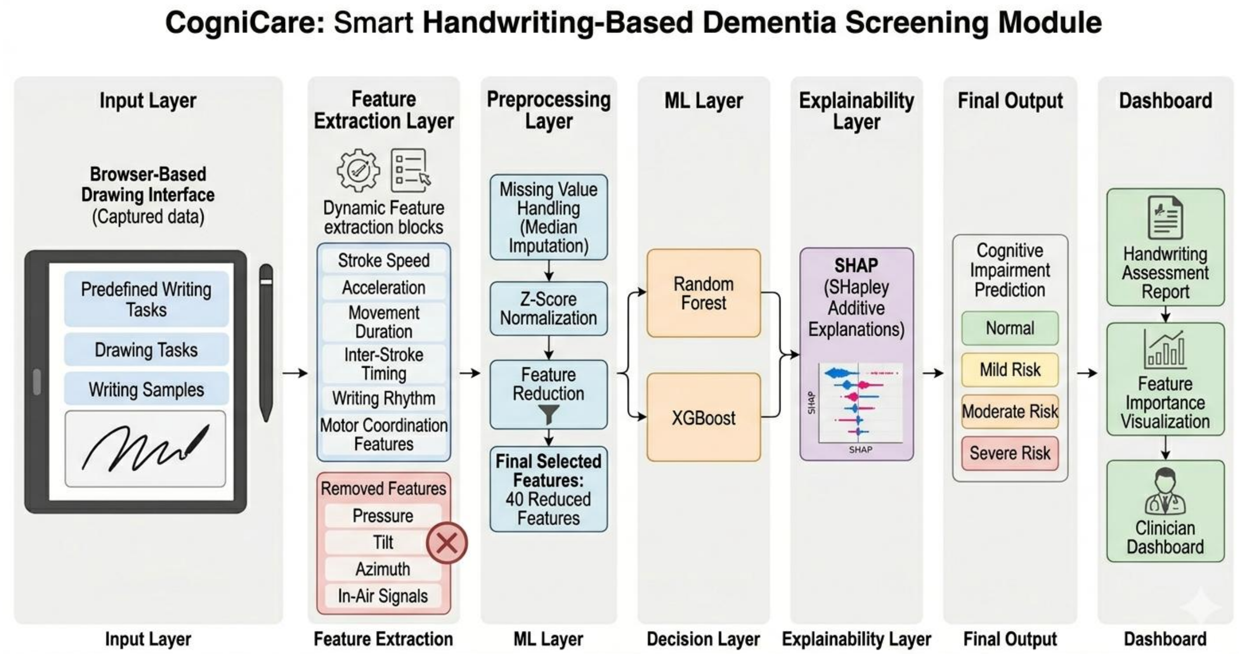


Figure 4 Handwriting-Based Cognitive Screening Module Architecture in CogniCare

3. Speech-Based Dementia Screening

The speech module analyzes guided speech responses to detect cognitive-linguistic impairment. Users complete three tasks: self-introduction, morning recall, and picture description. These tasks are selected to capture fluency, memory recall, and narrative coherence.

The speech pipeline includes both acoustic and transcript-based analysis. The acoustic pipeline uses Wav2Vec2-XLSR-300M and handcrafted acoustic features such as duration, tempo, ZCR, and RMS energy. The transcript pipeline extracts linguistic features such as word count, sentence count, speech rate, and lexical variation. Wav2Vec2, Random Forest, XGBoost, and a hybrid fusion model are used to generate task-level predictions and final speech risk scores.

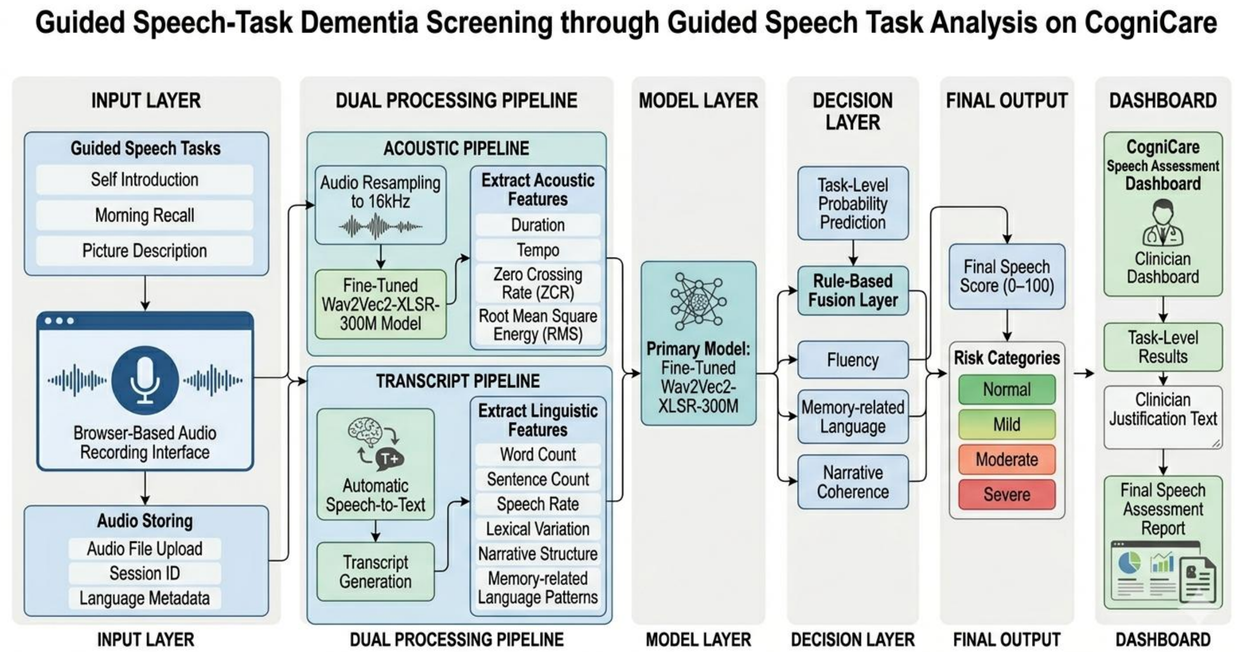


Figure 5 Guided Speech-Task Analysis Pipeline for Dementia Screening in CogniCare

4. MRI-Based Dementia Stage Classification

The MRI module is used as an optional advanced assessment when requested by the clinician or when screening results indicate the need for neuroimaging support. Brain MRI scans are analyzed using deep learning CNN models, including DenseNet-121, EfficientNet-B0, and ResNet-50. Their outputs are combined through a weighted ensemble to improve prediction robustness and reduce model-specific variance.

For interpretability, the module provides Grad-CAM++ heatmaps that highlight regions the model focused on during stage prediction. The MRI component supports four-class dementia staging (NonDemented, VeryMildDemented, MildDemented, ModerateDemented) and is designed as a clinical decision-support feature, not a mandatory first-line screening step.

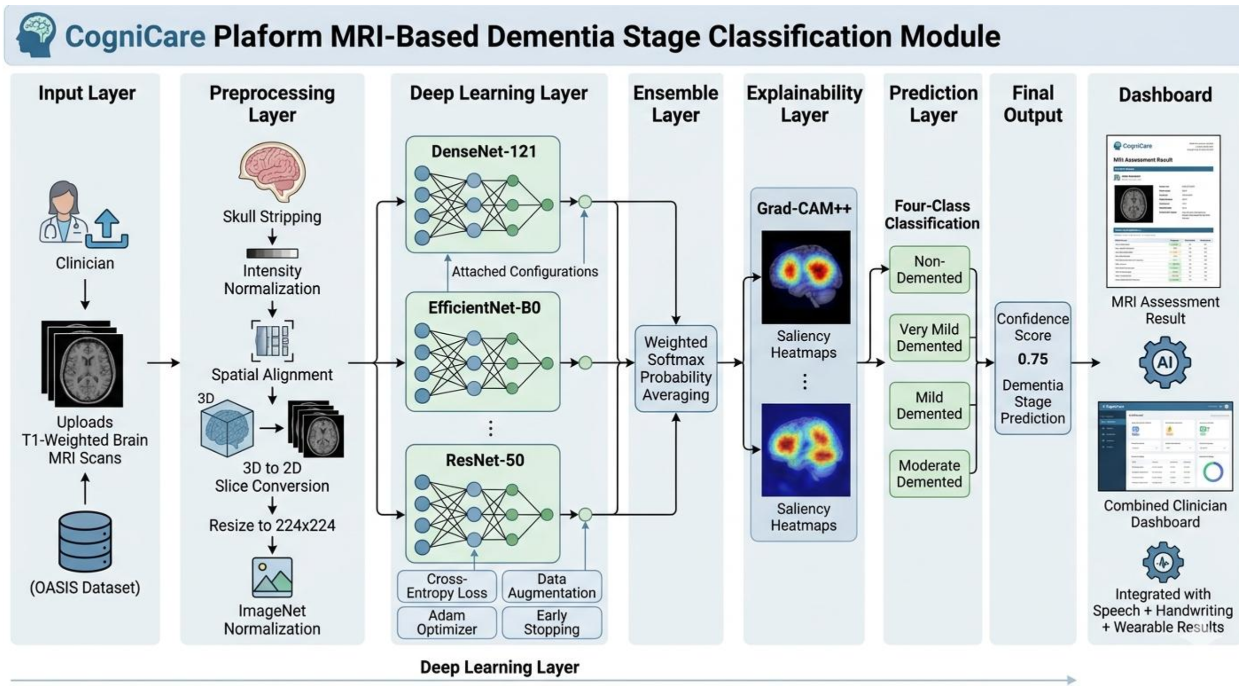


Figure 6 MRI-Based Dementia Stage Classification Module with Deep Learning and Explainability

2.1.4 Development Process

The development of CogniCare followed an iterative and modular approach. Since the system consists of multiple independent components, each module was designed, implemented, tested, and integrated progressively. This allowed the team to validate each component separately before combining them into the full platform.

The project was divided into core development streams: frontend development, backend gateway development, ML service implementation, database integration, clinician dashboard development, and report generation. GitHub was used for version control, while each member worked on assigned modules such as speech, handwriting, smart health, and MRI analysis.

The frontend was developed using React.js to provide a responsive browser-based interface. The backend was implemented using Node.js and Express.js to manage routes, sessions, authentication, prediction requests, and API communication. Machine learning models were implemented in Python and exposed as inference services. MongoDB was used for storing structured screening data and prediction results.

2.2. Commercialization Aspect of the Product

2.2.1 Problem–Solution Fit and Value Proposition

Dementia screening is often dependent on hospital visits, manual cognitive assessments, and specialist availability. This creates delays in early identification, especially in communities with limited access to neurological care. CogniCare addresses this issue by providing a digital, multimodal screening platform that can collect behavioral, physiological, and cognitive indicators through a web-based system.

The value of CogniCare lies in its ability to support early risk identification, reduce clinician workload, and provide interpretable screening reports. It does not replace clinical diagnosis; instead, it assists healthcare professionals in prioritizing patients who may need further evaluation.

2.2.2 Customer Segments and Stakeholders

The primary customers for CogniCare are hospitals, neurology clinics, elderly care centers, rehabilitation centers, and private healthcare providers. Public health organizations and NGOs working with elderly populations may also benefit from the platform.

The main users include elderly individuals, patients at risk of cognitive decline, caregivers, clinicians, and healthcare administrators. Clinicians benefit from structured reports, while caregivers and patients benefit from earlier awareness and follow-up recommendations.

2.2.3 Product Offering and Packaging

CogniCare can be offered as a cloud-based SaaS platform. A basic package may include smart health monitoring, handwriting analysis, speech screening, and report generation. An advanced package may include MRI stage classification, clinician dashboard tools, AI-assisted explanation, and institutional reporting features.

For hospitals with strict data governance requirements, a private cloud or on-premise deployment option can be provided.

2.2.4 Pricing Model

A practical pricing model would be based on subscription tiers. Clinics may pay monthly or annually based on the number of clinicians, patients, or screening sessions. Larger institutions can use enterprise pricing with custom deployment and support.

A possible pricing structure is:

Monthly Fee = Base Platform Fee + Number of Active Patients × Per-Patient Screening Fee

2.2.5 Competitive Advantage

CogniCare's main advantage is its multimodal and interpretable design. Many existing tools focus on only one input type, such as questionnaires, speech, handwriting, or MRI. CogniCare combines multiple indicators in one workflow and presents results in a clinician-facing dashboard.

Its browser-based design also improves accessibility because speech and handwriting tasks can be completed without specialized hardware. The modular structure allows future expansion into mobile screening, remote monitoring, and hospital system integration.

2.3. Testing and Implementation

To validate CogniCare, testing was carried out across functional, module, integration, system, and model evaluation levels. The purpose of testing was to ensure that each component worked independently and that the complete system could generate reliable screening outputs.

1. Functional Testing

Functional testing was performed on the main user workflows. These included user login, session creation, speech recording, handwriting task completion, lifestyle data entry, wearable data synchronization, prediction generation, clinician review, MRI upload, and final report generation.

Each workflow was tested to confirm that frontend inputs were correctly transmitted to the backend and that the system returned the expected outputs.

2. Module Testing

Each AI module was tested separately before system-level integration. The smart health module was tested using lifestyle, wearable, cognitive, and functional inputs. The handwriting module was tested using coordinate and timestamp data from the browser canvas. The speech module was tested using uploaded speech recordings and transcript-based feature extraction.

This ensured that each component produced valid intermediate outputs before being combined into the final report.

3. Integration Testing

Integration testing focused on communication between the React frontend, Node.js backend gateway, MongoDB database, and Python ML services. API endpoints were tested to verify data transfer between modules.

For the speech component, endpoints for session creation, audio upload, audio processing, final score generation, and assistant chat were tested. For handwriting and health modules, prediction requests and result storage were validated. Clinician dashboard integration was also tested to ensure that final results appeared correctly for medical review.

4. Model Evaluation

Machine learning models were evaluated using standard metrics such as accuracy, precision, recall, F1-score, and AUC. The smart health CatBoost model achieved strong performance with high accuracy and AUC. The handwriting module was evaluated using different feature configurations, and the reduced 40-feature Random Forest model was selected due to its balance between performance and efficiency. The speech module was evaluated using Wav2Vec2-XLSR-300M, Random Forest, XGBoost, and hybrid fusion models.

This model-level evaluation confirmed that each component could identify dementia-related patterns within its own modality.

5. System Testing

System testing verified whether CogniCare functioned as a complete screening platform. The full process from user input to final report generation was tested. This included checking whether data were stored correctly, predictions were retrieved successfully, and clinician reports were displayed with meaningful risk categories and explanations.

6. User Acceptance Testing

User acceptance testing focused on usability and clarity. The interface was reviewed for ease of navigation, clarity of instructions, readability of screening outputs, and usefulness of the clinician dashboard. Feedback was used to improve user flow, report wording, and presentation of screening results.

3. Results and Discussion

3.1. Smart Health Monitoring & Dementia Screening Risk Analysis

Results

The Smart Health Monitoring module was evaluated using a CatBoost gradient boosting classifier trained on a processed clinical dataset containing demographic, lifestyle, physiological, cognitive, and functional indicators. The model training incorporated preprocessing steps such as data cleaning, normalization, and feature engineering, while stratified cross-validation was applied to ensure robust and unbiased evaluation.

The experimental results demonstrate strong predictive performance across all evaluation metrics. The model achieved an **accuracy of 94.9%**, indicating a high proportion of correctly classified instances. In addition, the **precision, recall, and F1-score were all approximately 0.95**, reflecting a balanced performance in identifying both positive and negative cases. The model also achieved an **AUC score of 0.989**, indicating excellent discriminative capability between dementia and non-dementia classes.

The confusion matrix further confirms that the majority of samples were correctly classified, with minimal false positives and false negatives. These results highlight the effectiveness of the CatBoost model in handling heterogeneous healthcare data, including both numerical and categorical features.

Discussion

The results demonstrate that **lifestyle, physiological, and behavioral indicators provide strong predictive signals for dementia risk when analyzed using machine learning techniques**. The high accuracy and AUC values indicate that the model is capable of effectively distinguishing between different risk levels, making it suitable for early-stage cognitive screening.

One of the key strengths of this module is the use of the **CatBoost algorithm**, which is specifically designed to handle categorical variables efficiently without requiring extensive preprocessing. This makes it particularly well-suited for healthcare datasets, where features such as demographic information, lifestyle habits, and clinical attributes are often categorical in nature.

Another important aspect of this module is its ability to capture **long-term behavioral patterns**, such as physical activity levels, sleep behavior, and cognitive performance scores. Unlike traditional clinical assessments that rely on single-time evaluations, this approach provides a more comprehensive and realistic representation of an individual's cognitive health.

The integration of **wearable data (e.g., heart rate, step count, and activity levels)** further enhances the system's ability to support continuous monitoring. This enables the detection of subtle changes over time, which are often early indicators of cognitive decline.

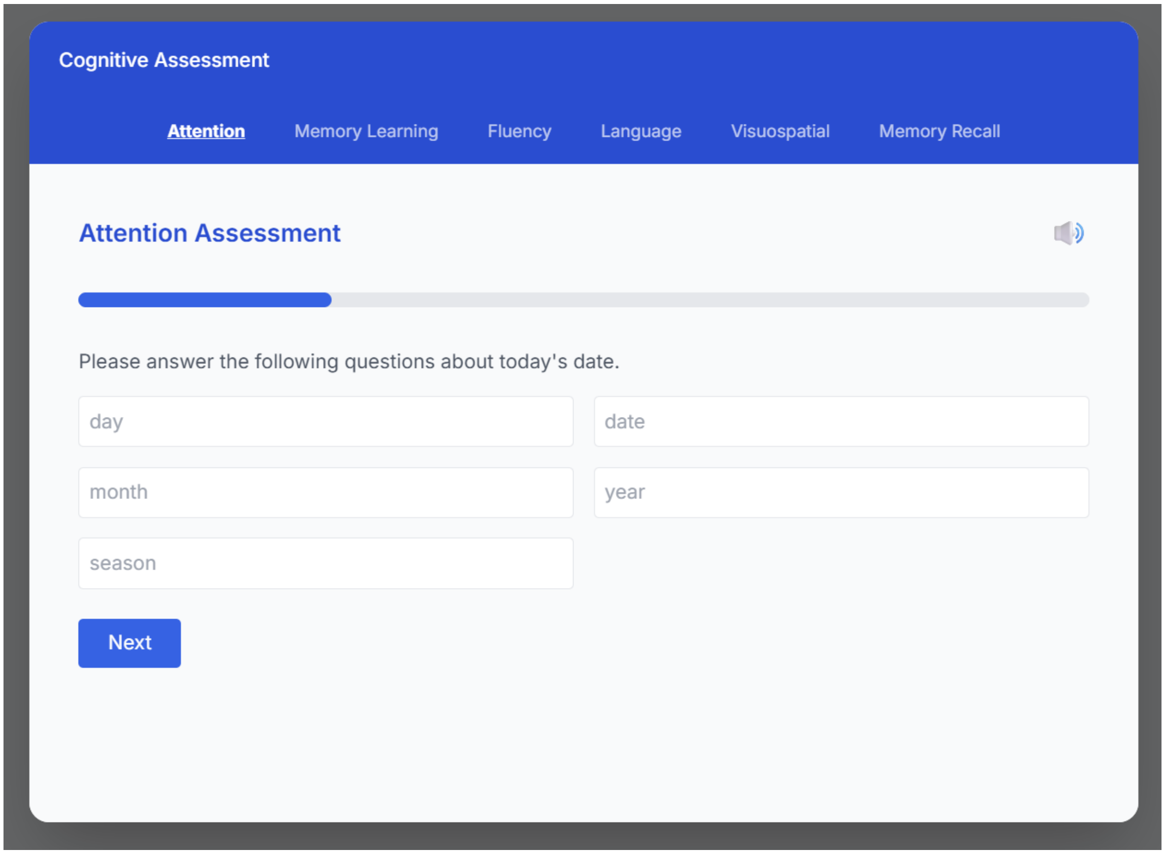
In addition, the use of a **rule-based fusion mechanism** to combine wearable predictions, cognitive assessment scores (ACE-III), and functional ability measures (IADL) improves interpretability. Rather than relying solely on a black-box prediction, the system provides structured outputs that can be understood and validated by clinicians.

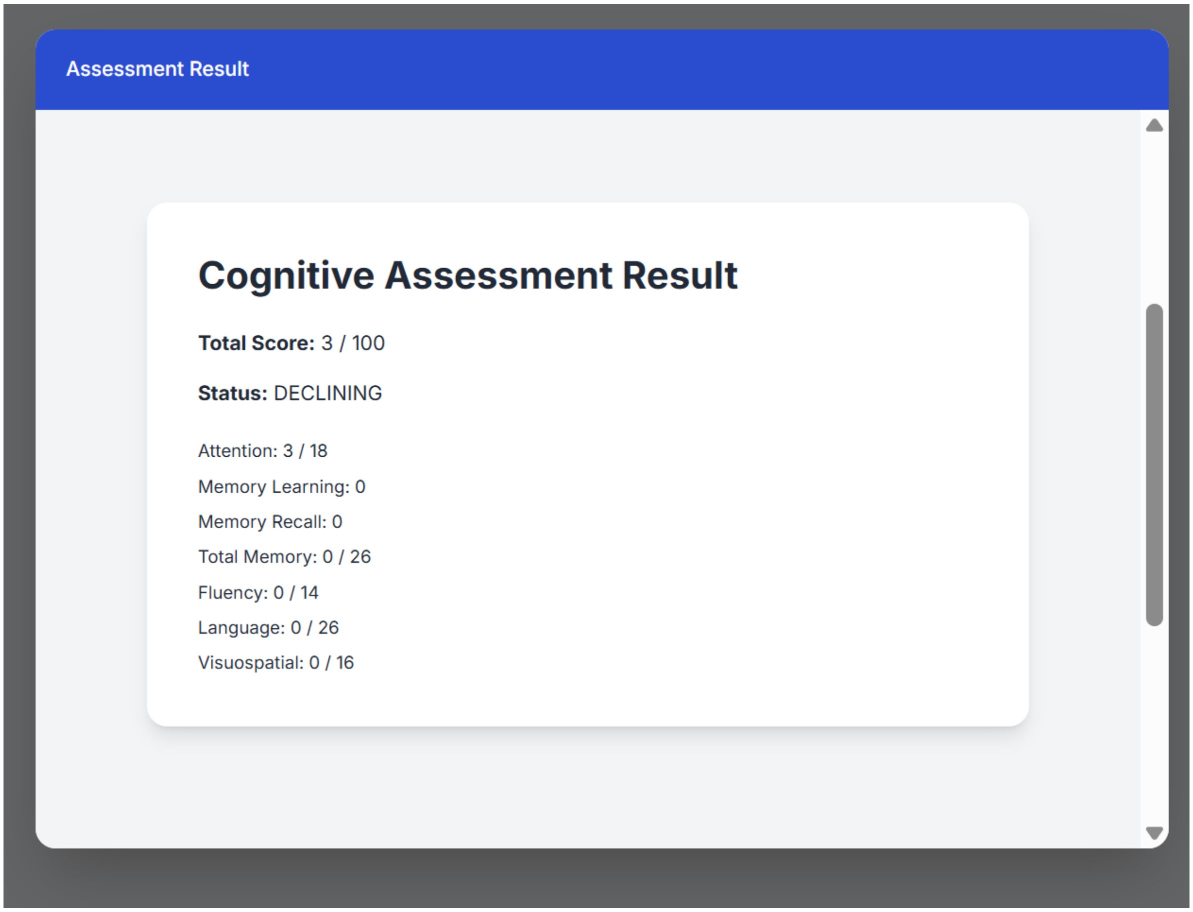
However, the model performance is influenced by the quality and diversity of the dataset. Since the training data is derived from controlled or semi-structured sources, further validation using real-world datasets is necessary to ensure generalizability across different populations.

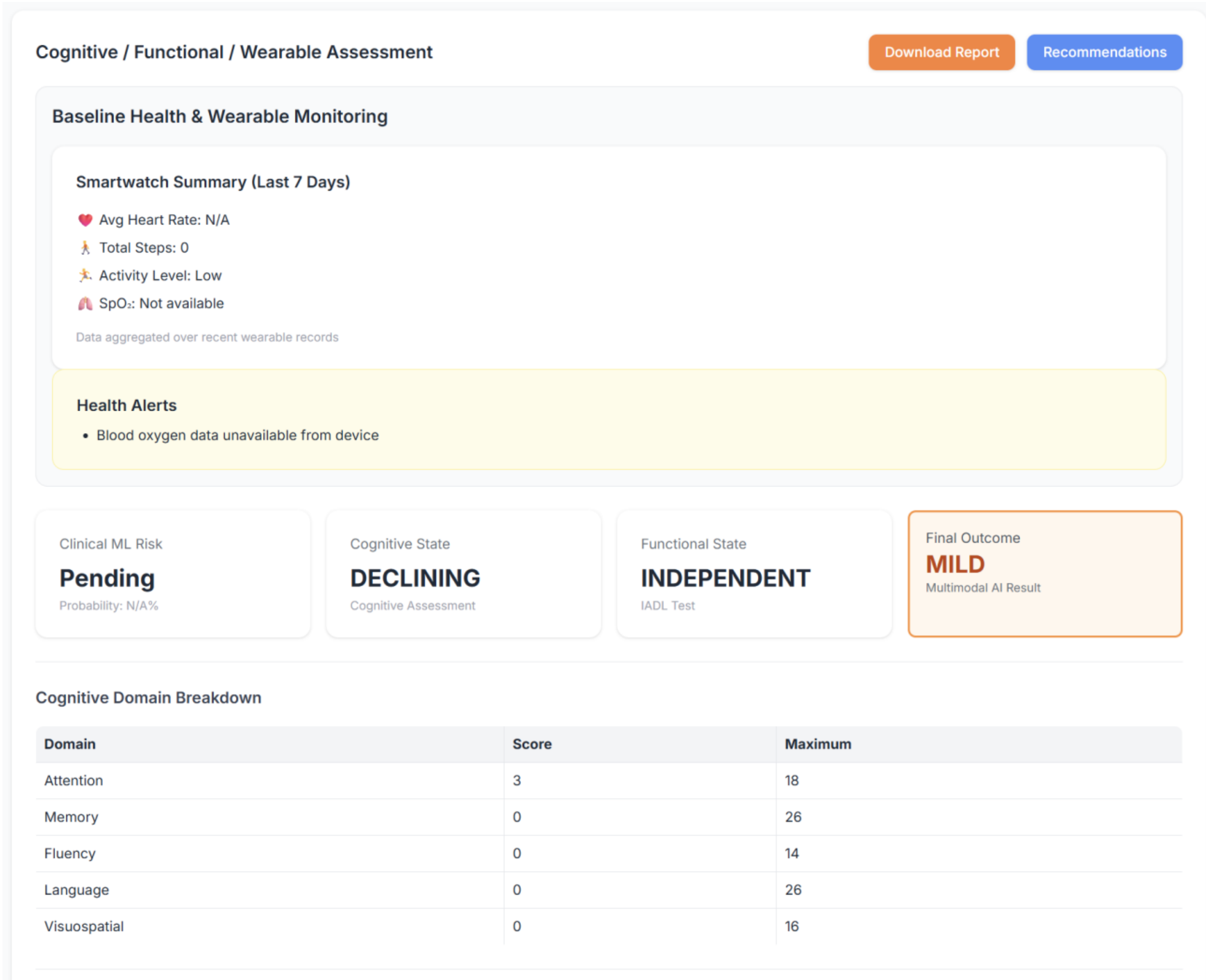
Overall, the Smart Health Monitoring module serves as a **core component of the CogniCare system**, providing a reliable, interpretable, and scalable approach to dementia risk prediction based on multimodal health data.

Metric	Value
Accuracy	0.949
Precision	0.95
Recall	0.95
F1 Score	0.95
AUC	0.989

Table 6 Performance Metrics of the CatBoost Model




Assessment Result
Cognitive Assessment Result
Total Score: 3 / 100
Status: DECLINING
Attention: 3 / 18
Memory Learning: 0
Memory Recall: 0
Total Memory: 0 / 26
Fluency: 0 / 14
Language: 0 / 26
Visuospatial: 0 / 16



3.2. Handwriting-Based Dementia Screening

Results

The handwriting-based dementia screening module was evaluated using multiple feature configurations and machine learning models, specifically Random Forest and XGBoost classifiers. The evaluation considered four feature sets: a full feature set (452 features), a browser-compatible set (96 features), a reduced set (40 features), and an ultra-reduced set (24 features).

The full feature model achieved the highest performance with an F1-score of approximately 0.886 and an AUC of 0.950. However, this configuration introduces high computational complexity and relies on sensor-dependent features such as pressure, tilt, and azimuth, which are not consistently available across devices.

The browser-compatible model (96 features) maintained strong performance with an F1-score of 0.869 and an AUC of 0.942, demonstrating improved feasibility for web-based environments. Further dimensionality reduction resulted in the 40-feature model achieving an F1-score of 0.870 and an AUC of 0.932, offering a strong balance between predictive performance and computational efficiency.

The ultra-reduced 24-feature model showed a slight performance drop (F1 = 0.838, AUC = 0.934), indicating that excessive feature reduction can lead to loss of discriminative information. Based on these results, the 40-feature configuration was selected as the optimal model for deployment.

Discussion

The experimental results demonstrate that **handwriting dynamics provide meaningful indicators of cognitive impairment**, particularly through motor-temporal features such as stroke velocity, acceleration, movement duration, and inter-stroke timing. These features capture underlying neurological and motor coordination changes associated with early-stage dementia.

A key contribution of this module is the identification of a **reduced feature set that maintains strong predictive performance while significantly improving efficiency**. By removing sensor-dependent attributes such as pressure, tilt, and in-air signals, the system becomes more robust and compatible with browser-based data collection environments.

This design enhances **device independence and scalability**, allowing the system to function across a wide range of input devices without requiring specialized hardware. Furthermore, the reduced computational complexity enables faster processing, making the module suitable for real-time or near real-time screening applications.

In addition, the integration of **Explainable AI techniques (SHAP)** provides transparency by identifying the most influential handwriting features contributing to model predictions. This improves interpretability and supports clinical trust in the system.

Overall, the handwriting module serves as a **lightweight, non-invasive, and scalable behavioral biomarker**, contributing valuable insights to the multimodal dementia screening framework while maintaining practical deployment feasibility.

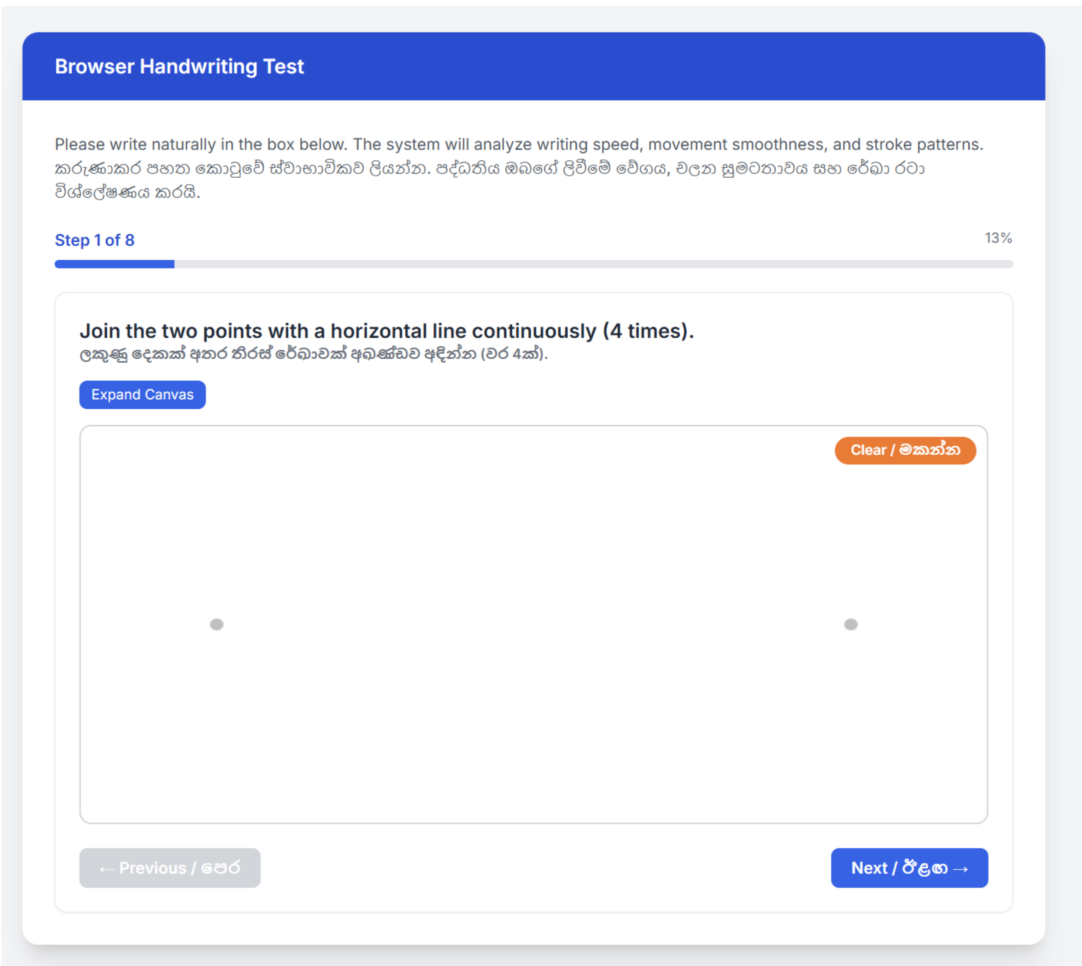
Model Version	No. of Features	Best F1 Score	AUC	Interpretation
Full Model	452	0.886	0.950	Strongest overall performance
Browser Model	96	0.869	0.942	Very close to full model
Reduced Model	40	0.870	0.932	Comparable to 96-feature model

Ultra Reduced Model	24	0.838	0.934	Noticeable performance drop

Figure 7 Performance comparison of machine learning models across different feature configurations

Feature Set	No. of Features	Models Trained	Best Model	Best F1 Score
Full Model	452	Random Forest, XGBoost	XGBoost	0.886
Browser Model	96	Random Forest, XGBoost	XGBoost	0.869
Reduced Model	40	Random Forest, XGBoost	Random Forest	0.870
Ultra Reduced Model	24	Random Forest, XGBoost	Random Forest	0.838

Figure 8 Summary of model selection across feature sets and corresponding best-performing algorithms.



Browser Handwriting Test

Please write naturally in the box below. The system will analyze writing speed, movement smoothness, and stroke patterns.
කරුණාකර පහත කොටුවේ ස්වාභාවිකව ලියන්න. පද්ධතිය ඔබගේ ලිවීමේ වේගය, චලන සුමටතාවය සහ රේඛා රටා විශ්ලේෂණය කරයි.

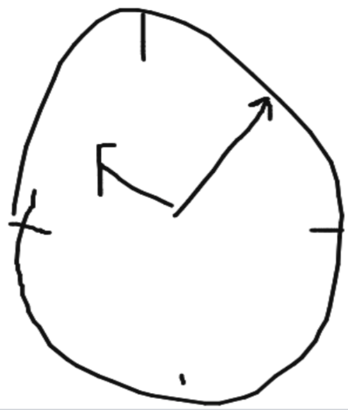
Step 8 of 8 100%

Draw a clock showing 10:10.

10:10 වේලාව පෙන්වන ඔරලෝසුවක් අඳින්න.

Expand Canvas

Clear / මකන්න



← Previous / පෙර

Submit Assessment

Browser Handwriting Screening Result / බ්‍රවුසරය හරහා ලිවීමේ පරීක්ෂණ ප්‍රතිඵල

Healthy Pattern / සාමාන්‍ය ලිවීමේ රටාව

Dementia Risk Probability / ඩිමෙන්ශියා අවදානම් ප්‍රතිශතය: 46.3%

This prediction was generated using handwriting motor dynamics extracted directly from the browser.
මෙම ප්‍රතිඵලය බ්‍රවුසරයෙන් ලබාගත් ලිවීමේ චලන තොරතුරු භාවිතයෙන් ගණනය කර ඇත.

Top Contributing Features (SHAP) / ප්‍රධාන බලපාන ලක්ෂණ

Writing smoothness / ලිවීමේ සුමටතාවය (Task 24) Towards Dementia / ඩිමෙන්ශියා දෙසට බලපායි	0.0748
Average writing acceleration / සාමාන්‍ය ලිවීමේ වේග වෙනස්වීම (Task 24) Towards Dementia / ඩිමෙන්ශියා දෙසට බලපායි	0.0626
Time taken to complete task / කාර්යය සම්පූර්ණ කිරීමට ගත වූ කාලය (Task 2) Towards Healthy / සාමාන්‍ය තත්වයට බලපායි	-0.0292
Writing speed / ලිවීමේ වේගය (Task 9) Towards Dementia / ඩිමෙන්ශියා දෙසට බලපායි	0.0284
Average writing acceleration / සාමාන්‍ය ලිවීමේ වේග වෙනස්වීම (Task 21) Towards Healthy / සාමාන්‍ය තත්වයට බලපායි	-0.0268
Time taken to complete task / කාර්යය සම්පූර්ණ කිරීමට ගත වූ කාලය (Task 24) Towards Healthy / සාමාන්‍ය තත්වයට බලපායි	-0.0255
Stroke movement consistency / රේඛා චලනයේ ස්ථාවරතාවය (Task 9) Towards Healthy / සාමාන්‍ය තත්වයට බලපායි	-0.0241
Time taken to complete task / කාර්යය සම්පූර්ණ කිරීමට ගත වූ කාලය (Task 8) Towards Healthy / සාමාන්‍ය තත්වයට බලපායි	-0.0231
Time taken to complete task / කාර්යය සම්පූර්ණ කිරීමට ගත වූ කාලය (Task 9) Towards Healthy / සාමාන්‍ය තත්වයට බලපායි	-0.0209
Writing speed / ලිවීමේ වේගය (Task 5) Towards Dementia / ඩිමෙන්ශියා දෙසට බලපායි	0.0204
Writing speed / ලිවීමේ වේගය (Task 8) Towards Dementia / ඩිමෙන්ශියා දෙසට බලපායි	0.0204
Writing smoothness / ලිවීමේ සුමටතාවය (Task 21) Towards Healthy / සාමාන්‍ය තත්වයට බලපායි	-0.0195

3.3. Dementia screening through guided speech-task analysis

Results

The speech-based dementia screening module was evaluated using multiple modeling approaches to capture both acoustic and linguistic characteristics of speech. The evaluated models include a deep learning model (Wav2Vec2-XLSR-300M) for raw audio classification, an acoustic feature-based model using Random Forest, a transcript-based model using XGBoost, and a hybrid fusion model combining deep audio embeddings with linguistic features.

The experimental results demonstrate that the **Wav2Vec2-XLSR-300M model** achieved strong performance, with an accuracy of **0.79**, precision of **0.80**, recall of **0.77**, F1-score of **0.78**, and an AUC of **0.85**. The **acoustic feature-based Random Forest model** showed comparatively lower performance, achieving an accuracy of **0.73** and an AUC of **0.79**, indicating that handcrafted acoustic features alone are less effective in capturing complex speech patterns.

Similarly, the **transcript-based XGBoost model** achieved moderate performance, with an accuracy of **0.76** and an AUC of **0.84**, demonstrating the contribution of linguistic features such as word count, speech rate, and lexical variation in identifying cognitive impairment.

Discussion

The results highlight the importance of **multimodal speech representation** in detecting cognitive impairment. While deep learning models such as Wav2Vec2-XLSR-300M effectively capture complex acoustic patterns, they may not fully represent higher-level linguistic structures such as coherence, vocabulary richness, and memory-related language organization.

The acoustic feature-based model, although useful, showed lower performance due to its reliance on handcrafted features such as pitch, tempo, and energy, which may not sufficiently capture subtle cognitive changes. On the other hand, the transcript-based model demonstrates that **linguistic features provide meaningful indicators of cognitive decline**, particularly in areas such as memory recall, fluency, and narrative structure.

The superior performance of the hybrid model confirms that **combining deep acoustic representations with linguistic features results in a more robust and comprehensive model**. This aligns with the understanding that dementia affects both speech production and language processing, requiring analysis at multiple levels.

Another key strength of this module is the use of **guided speech tasks**, including self-introduction, memory recall, and picture description, which are designed to capture different cognitive domains. This structured approach ensures consistency in data collection while enabling the system to evaluate multiple aspects of cognitive functioning.

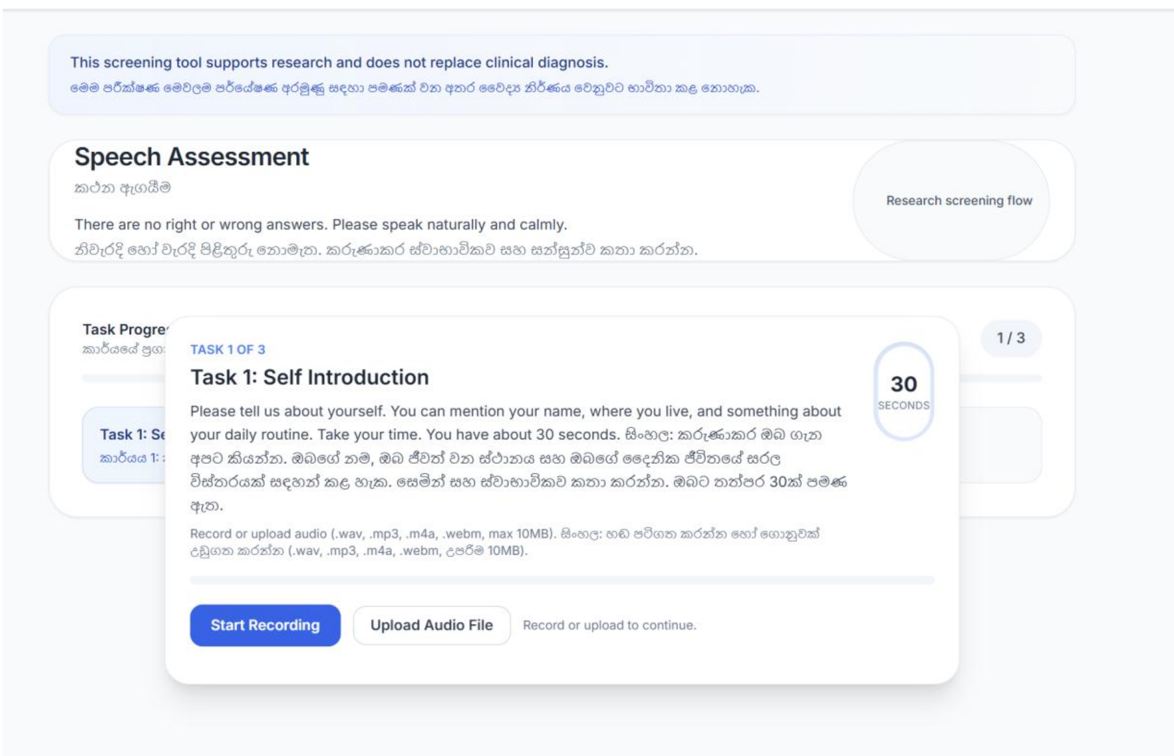
Furthermore, the use of a **rule-based fusion layer** to map model outputs into clinically meaningful categories (e.g., Normal, Mild, Moderate, Severe) enhances interpretability and usability in real-world settings.

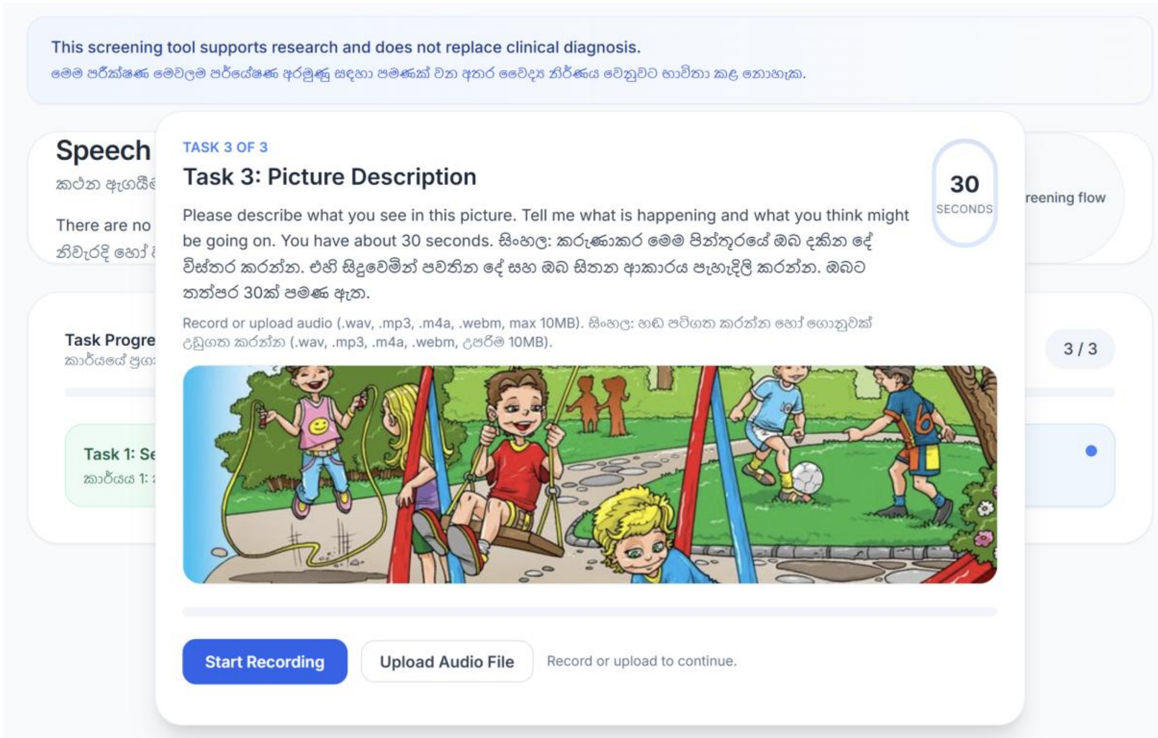
However, the performance of the speech module may be influenced by factors such as background noise, recording quality, and language variability. Future improvements could include noise-robust models, multilingual support, and larger real-world datasets to improve generalizability.

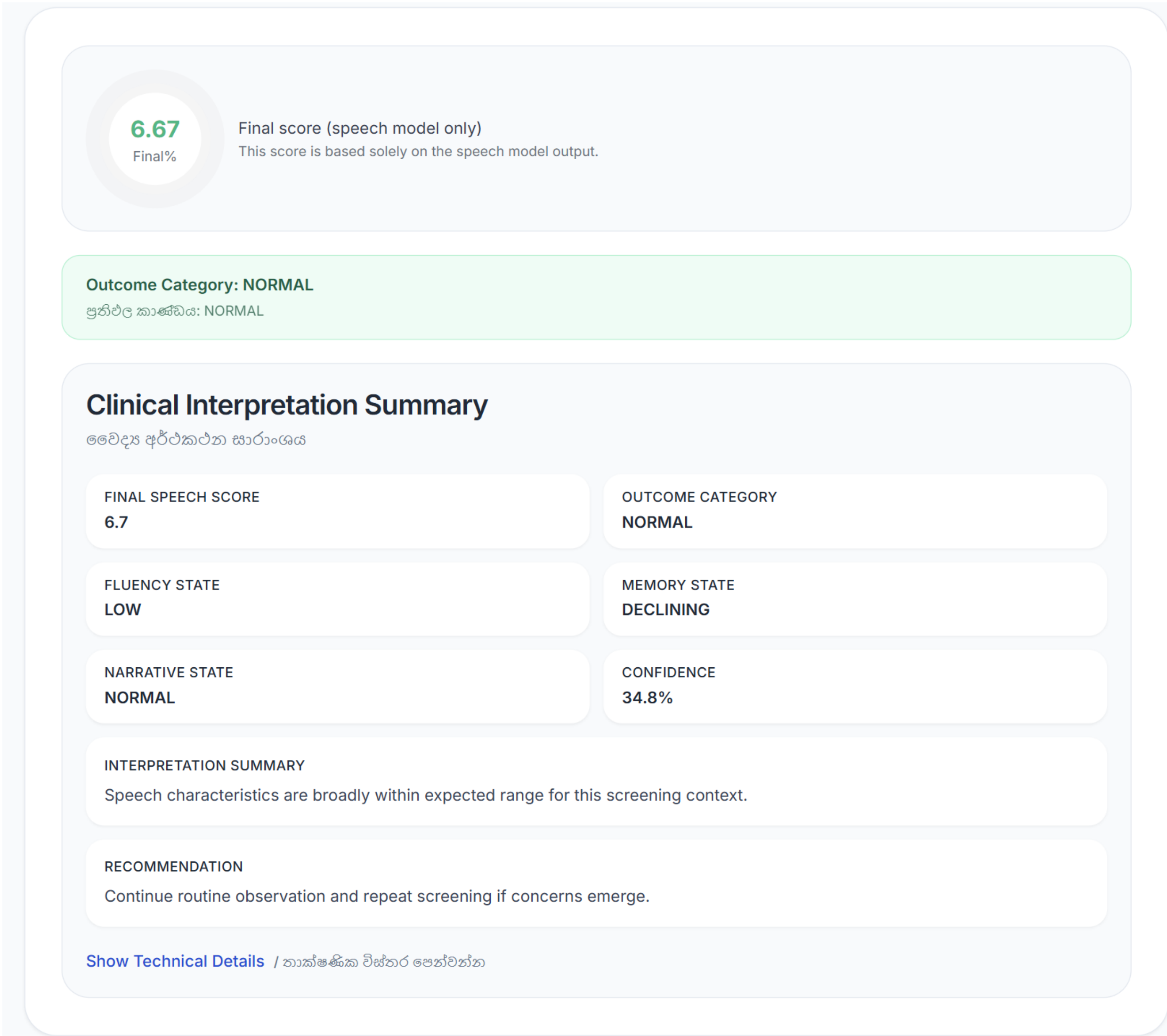
Overall, the speech module provides a **scalable, non-invasive, and behaviorally rich approach to dementia screening**, complementing other modalities within the CogniCare system.

Model	Accuracy	Precision	Recall	F1 Score	AUC
Wav2Vec2-XLSR-300M	0.79	0.80	0.77	0.78	0.85
Acoustic Model (Random Forest)	0.73	0.74	0.70	0.71	0.79
Transcript Model (XGBoost)	0.76	0.77	0.74	0.75	0.84

Table 7 Performance Metrics of Speech-Based Models




This screening tool supports research and does not replace clinical diagnosis.
මෙම පරීක්ෂණ මෙවලම පර්යේෂණ අරමුණු සඳහා පමණක් වන අතර වෛද්‍ය නිර්ණය වෙනුවට භාවිතා කළ නොහැක.
Speech
TASK 3 OF 3
Task 3: Picture Description
30
SECONDS
Please describe what you see in this picture. Tell me what is happening and what you think might be going on. You have about 30 seconds. සිංහල: කරුණාකර මෙම පින්තූරයේ ඔබ දකින දේ විස්තර කරන්න. එහි සිදුවෙමින් පවතින දේ සහ ඔබ සිතන ආකාරය පැහැදිලි කරන්න. ඔබට තත්පර 30ක් පමණ ඇත.
Record or upload audio (.wav, .mp3, .m4a, .webm, max 10MB). සිංහල: හඬ පටිගත කරන්න හෝ ගොනුවක් උඩුගත කරන්න (.wav, .mp3, .m4a, .webm, උපරිම 10MB).
3 / 3
Start Recording
Upload Audio File
Record or upload to continue.



3.4. MRI-Based Dementia Stage Classification

Results

The MRI-based dementia classification module was evaluated using multiple deep learning architectures to predict dementia stages from structural brain MRI images. The models considered include **DenseNet-121, EfficientNet-B0, and ResNet-50**, each trained on preprocessed MRI data and evaluated using standard classification metrics.

Among the individual models, **DenseNet-121** achieved the highest performance, with an **accuracy of 0.9484**, **macro F1-score of 0.9657**, **macro precision of 0.9633**, and **macro recall of 0.9684**, indicating strong capability in distinguishing between different dementia stages.

The **EfficientNet-B0** model achieved moderate performance, with an accuracy of **0.8562** and macro F1-score of **0.9067**, while **ResNet-50** showed slightly lower performance with an accuracy of **0.8406** and macro F1-score of **0.8921**.

To further enhance prediction reliability, an **ensemble model** was developed by combining the outputs of the three architectures using a weighted averaging approach. The ensemble model achieved an **accuracy of 0.9375**, **macro F1-score of 0.9606**, **macro precision of 0.9591**, and **macro recall of 0.9636**.

These results demonstrate that while DenseNet-121 performs best individually, the ensemble model provides more **balanced and stable performance across all classes**.

Discussion

The results indicate that **deep learning models are highly effective in capturing structural brain patterns associated with different stages of dementia**. The strong performance of DenseNet-121 can be attributed to its dense connectivity structure, which enables efficient feature reuse and improved gradient flow, making it particularly suitable for medical image analysis.

Although DenseNet-121 achieved the highest individual accuracy, the ensemble model demonstrated improved **robustness and generalization**. By combining predictions from multiple architectures, the ensemble approach reduces model-specific biases and enhances the system's ability to perform consistently across different input variations.

The use of **multiple CNN architectures** allows the system to learn diverse feature representations from MRI data, including spatial patterns related to brain atrophy and structural degeneration. This contributes to more reliable stage classification compared to relying on a single model.

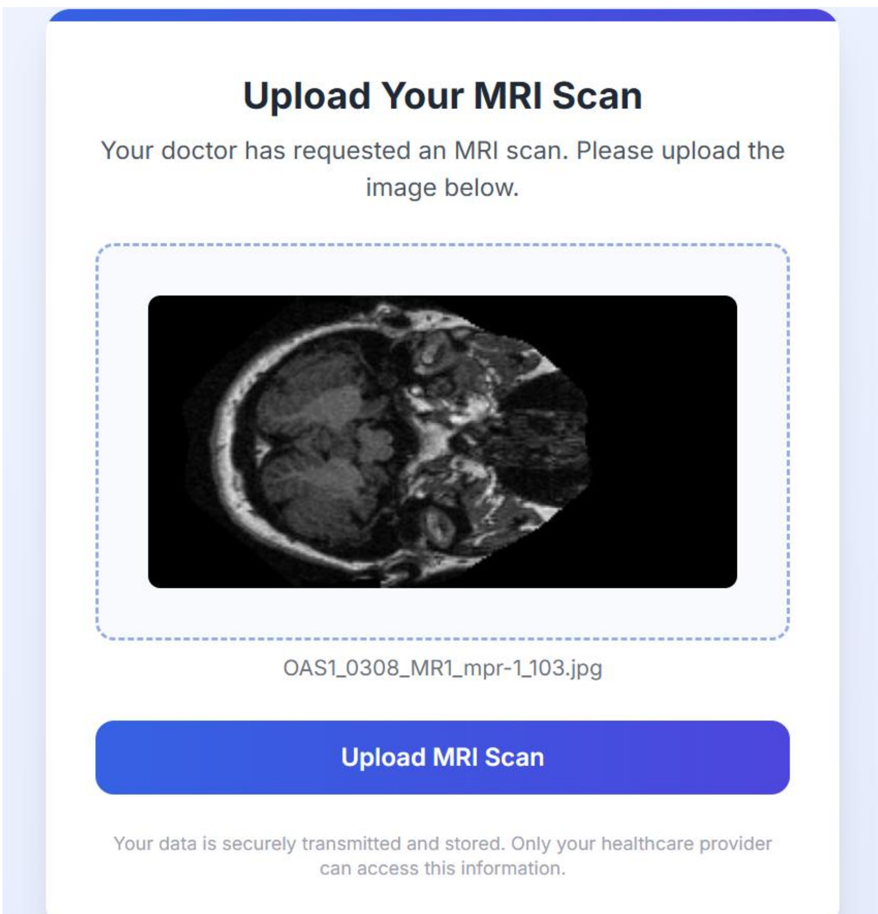
Another key strength of this module is the integration of **Explainable AI techniques such as Grad-CAM++**, which generate visual heatmaps highlighting the brain regions influencing the model's predictions. This improves interpretability and provides clinicians with insights into the underlying reasoning of the model, increasing trust in the system.

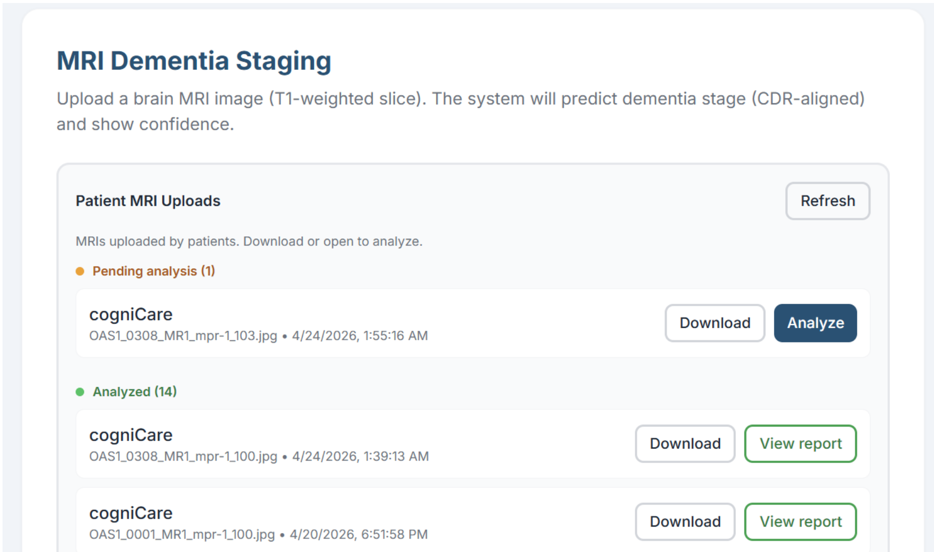
However, the MRI-based approach has certain limitations. The performance is dependent on the availability and quality of labeled MRI datasets, and the preprocessing steps (such as skull stripping and normalization) may introduce variability. Additionally, MRI analysis requires specialized imaging equipment, making it less accessible compared to behavioral screening methods.

Despite these limitations, the MRI module serves as a **high-accuracy, clinically valuable component** within the CogniCare system, particularly for advanced assessment scenarios. When combined with other modalities such as speech, handwriting, and lifestyle data, it significantly enhances the overall reliability of dementia risk screening.

Model	Accuracy	Macro F1	Macro Precision	Macro Recall
DenseNet-121	0.9484	0.9657	0.9633	0.9684
EfficientNet-B0	0.8562	0.9067	0.9085	0.9171
ResNet-50	0.8406	0.8921	0.8957	0.8889
Ensemble Model	0.9375	0.9606	0.9591	0.9636

Table 8 Performance of MRI-Based Dementia Classification Models




MRI Dementia Staging
Upload a brain MRI image (T1-weighted slice). The system will predict dementia stage (CDR-aligned) and show confidence.
Patient MRI Uploads
Refresh
MRIs uploaded by patients. Download or open to analyze.
Pending analysis (1)
cogniCare
OAS1_0308_MR1_mpr-1_103.jpg • 4/24/2026, 1:55:16 AM
Download
Analyze
Analyzed (14)
cogniCare
OAS1_0308_MR1_mpr-1_100.jpg • 4/24/2026, 1:39:13 AM
Download
View report
cogniCare
OAS1_0001_MR1_mpr-1_100.jpg • 4/20/2026, 6:51:58 PM
Download
View report

MRI Assessment Result

Very Mild Demented

Confidence: 54.1%

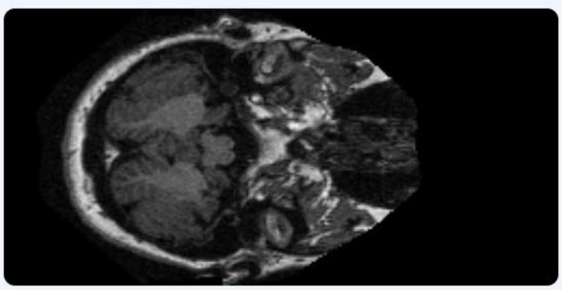
Early signs detected. Clinical follow-up recommended.

Note: card color indicates risk severity; heatmap red indicates model attention regions, not severity.

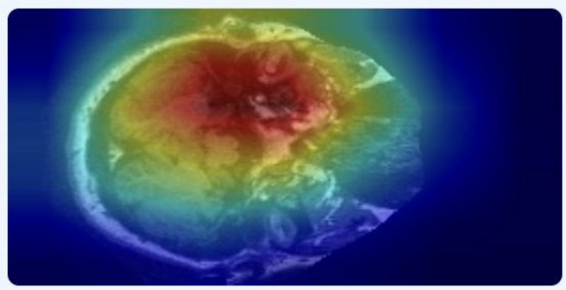
CLASS PROBABILITIES

Stage	Probability
MildDemented	43.4%
ModerateDemented	0.1%
NonDemented	2.4%
VeryMildDemented	54.1%

ORIGINAL MRI SCAN



EXPLAINABILITY (GRAD-CAM++)



Hide Heatmap

Brain Region Analysis

37.4%	39.2%	23.4%	57
High Attention	Medium Attention	Low Attention	Severity (0-100)

Interpretation: High attention regions (red/yellow in heatmap) are areas the model focused on when making its prediction. For dementia cases, these typically correspond to structural changes like ventricular enlargement or cortical thinning.

Your MRI Comment

Add your comments on the MRI result...

4. Summary of Each Student's Contribution

IT Number	Name	Component	Role
IT22358998	Senaratne H D	Handwriting-Based Dementia Screening Module	**Team Leader** Developed the handwriting-based dementia screening module, including data preprocessing, feature extraction, machine learning model development, and explainability integration. Led the project as team leader, coordinated with the external supervisor, and managed real-world testing with elderly participants.
IT22901644	Dissanayake M.P.S.	MRI-Based Dementia Stage Classification	Developed the MRI-based dementia staging module using DenseNet-121, EfficientNet-B0, and ResNet-50 with weighted ensemble prediction. Implemented MRI preprocessing/validation, integrated explainable AI (Grad-CAM, Grad-CAM++, Score-CAM, Fusion CAM), built backend APIs and frontend result/report views, and supported clinical pilot validation with real patient MRI scans.

IT22358820	Piruthivi K	SMART HEALTH MONITORING AND DEMENTIA RISK ANALYSIS	Developed wearable-based dementia risk prediction system using CatBoost, implemented feature engineering, cognitive & functional assessment integration, and rule-based decision mechanism.
IT22888952	Natheer H A H	Speech-Based Dementia Screening	Contributed to the development of the speech-based dementia screening module and supported the overall system implementation as a team member. Handled frontend development tasks. Developed backend API integration. Built the explainable AI assistant for clinician-friendly result interpretation. Maintained team communication and coordination through regular discussions and calls, while supporting testing, deployment preparation, and overall system validation.

Table 1: Summary Of Each Student's Contribution

5. Conclusions

This study presented CogniCare, a multimodal AI-assisted dementia screening platform that integrates wearable health monitoring, cognitive and functional assessments, handwriting analysis, speech evaluation, and MRI-based stage classification within a unified framework. The system demonstrates the effectiveness of combining multiple data sources to improve the reliability and interpretability of dementia risk screening.

Experimental results across the individual components show strong performance. The Smart Health Monitoring module achieved high predictive accuracy using a CatBoost model with an AUC of 0.989 and 94.9% accuracy. The handwriting analysis module demonstrated that writing dynamics can effectively indicate cognitive impairment, achieving an F1-score of 0.870. For neuroimaging analysis, deep learning models including DenseNet-121, EfficientNet-B0, and ResNet-50 were combined in an ensemble approach, achieving 93.75% accuracy with a Macro F1-score of 0.9606.

By integrating outputs from multiple modalities, the proposed system enhances screening robustness and provides interpretable results that support clinicians in early dementia risk assessment. Overall, CogniCare demonstrates the potential of combining artificial intelligence with digital health technologies to enable scalable, accessible, and effective cognitive screening, while ensuring that final diagnostic decisions remain under professional medical supervision.

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